

SUBJECT Math **TOPIC** Sum of angles in a triangle **CROSS-CURRICULAR** Origami

CONTEXT 9th grade. Only paper available.

 Narrative Hook

A master origami artist challenges her students: "Every crane, every fox, every flower — they all hide the same secret inside their folds. Today, you won't just fold paper... you'll *prove* one of geometry's oldest truths using nothing but your hands." Students look down at a single sheet of paper and wonder — how can folding reveal a mathematical law?

 Timeline

PHASE	ACTIVITY	DURATION
1	Narrative hook & framing the question	5 min
2	Guided folding — creating triangles	10 min
3	Tearing & arranging corners (discovery)	10 min
4	Class discussion & formalizing the theorem	10 min
5	Applying to a simple origami base fold	10 min
6	Reflection & journaling	5 min
Total	50 min	

 Materials

- Plain paper (at least 3 sheets per student — no scissors, no rulers needed)
- Pencil or pen (for labeling angles)
- Optional: colored paper for the origami extension

Step-by-Step Instructions

- 1. Hook (5 min):** Read the narrative hook aloud or display it. Ask students: *"Do you think all triangles follow the same angle rule — even oddly shaped ones?"* Record predictions on the board.
- 1. Fold a triangle (5 min):** Have each student fold a sheet of paper to create a large, irregular triangle. Emphasize they should all make *different* triangles — some wide, some narrow, some nearly right-angled. Label the three interior angles **A**, **B**, and **C** lightly in pencil.
- 1. Tear the corners (5 min):** Students carefully tear off each labeled corner of the triangle, keeping the corner piece intact. They now have three separate angle-pieces labeled A, B, and C.
- 1. Arrange along a straight line (5 min):** Students place the torn corners side-by-side along the straight edge of a folded paper strip, vertices touching, with no gaps or overlaps. Observe: the three angles form a perfectly flat, straight line.
- 1. Connect to the theorem (10 min):** Guide the class: *"A straight line measures 180°. What does this tell us about $A + B + C$?"* Formalize on the board: $\angle A + \angle B + \angle C = 180^\circ$. Discuss why this worked for every student's triangle — even though they were all different shapes.
- 1. Origami connection (10 min):** Demonstrate a simple origami **preliminary square base**: fold paper diagonally corner-to-corner. Ask students to identify the triangles formed and verify the angle sum using their theorem. Pose the question: *"When an origami fold creates a crease that bisects an angle, what does that tell us about the two new angles?"*
- 1. Reflection & journaling (5 min):** Students respond to the reflection questions below independently before a brief class share-out.

Reflection Questions

1. Your classmates all folded different triangles, yet you all got the same result. What does that tell you about the nature of mathematical proof?
2. When you folded the paper diagonally to make the origami base, what types of triangles appeared? What were their angles?
3. Could this paper-tearing method be considered a *proof*? What are its limitations compared to a formal geometric proof?
4. Origami artists use angle relationships to make precise, repeatable folds. Can you think of another profession or craft where understanding triangle angles would be essential?
5. If you tried this on a sphere (like a globe) instead of flat paper, do you think the angle sum would still be 180° ? Why or why not?

Assessment Rubric

CRITERION	DEVELOPING (1)	PROFICIENT (2)	EXEMPLARY (3)
Conceptual Understanding	Cannot explain why the angles sum to 180°	Explains the result using the paper activity but struggles with formal language	Clearly articulates the theorem and connects it to the straight-line observation
Mathematical Reasoning	Does not connect the physical activity to the angle-sum theorem	Makes the connection with teacher prompting	Independently generalizes the result and can apply it to new triangle examples
Origami–Geometry Connection	Cannot identify triangles or angles in the origami fold	Identifies triangles but cannot state their angle properties	Accurately names angles in the origami base and applies the theorem to reason about them
Reflection Quality	Responses are one word or off-topic	Responses show basic understanding but lack elaboration	Responses demonstrate critical thinking, curiosity, and personal connection to the concepts

Customization Zones

1. **Swap the triangle type for a targeted challenge** — For advanced students, have them fold a right triangle specifically and verify that the two acute angles must sum to exactly 90° .

 [INSERT TEACHER NOTE HERE]

1. **Add a visual anchor for the discovery moment** — Display a photograph of the torn corners arranged along the straight edge so students have a reference before they begin.

 [INSERT PHOTO OR IMAGE HERE]

1. **Provide a structured recording sheet** — Give students a worksheet to sketch their triangle, label the angles, record their prediction, and write the conclusion in their own words.

 [INSERT STUDENT WORKSHEET HERE]

1. **Differentiate the origami complexity** — For students who finish early, introduce the **bird base** or **waterbomb base** and ask them to map *all* the triangles they can find within the fold diagram.

 [INSERT TEACHER NOTE HERE]